If ρ has the form $\rho = \exp(-\beta\mu_{\mathrm{P}}\eta^2/2) \times \psi(\boldsymbol{q},\boldsymbol{p},V,\xi)$ then eliminate η, reducing $\mathcal{L}^\dagger\rho = 0$ to

$$-\nabla_{\boldsymbol{q}} \cdot (\boldsymbol{q}\psi) + \nabla_{\boldsymbol{p}} \cdot (\boldsymbol{p}\psi) - \frac{\partial}{\partial V}(V\psi) + \beta(V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \mathrm{P}V + \mathrm{k_B T})\psi = 0.$$

This results in

$$-\boldsymbol{q} \cdot \nabla_{\boldsymbol{q}}\psi + \boldsymbol{p} \cdot \nabla_{\boldsymbol{p}}\psi - \psi - V\frac{\partial\psi}{\partial V} + (\beta V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \beta\mathrm{P}V + 1)\psi = 0. \tag{8.74}$$

A solution (the desired one) is given by

$$\psi \propto \exp(-\beta(H + \mathrm{P}V)),$$

where substituting into (8.74) gives

$$\Big[\beta\boldsymbol{q} \cdot \nabla U(\boldsymbol{q}) - \beta\boldsymbol{p} \cdot \boldsymbol{M}^{-1}\boldsymbol{p} - 1 + \beta V\frac{\partial U}{\partial V} + \beta\mathrm{P}V + (\beta V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \beta\mathrm{P}V + 1)\Big]\psi = 0,$$

which is quickly checked to be true, and hence the dynamics preserves $\rho = \exp(-\beta\mu_{\mathrm{P}}\eta^2/2) \times \psi$.

It is difficult to see how their could be other solutions ψ, but we know from the Nosé-Hoover case that this must be possible. We improve the robustness of the method by introducing Ornstein-Uhlenbeck type stochastic processes in the two auxiliary variables ξ and η, yielding the system

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p} + \eta\boldsymbol{q}, \tag{8.75}$$

$$\dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) - \eta\boldsymbol{p} - \xi\boldsymbol{p}, \tag{8.76}$$

$$\dot{V} = \eta V, \tag{8.77}$$

$$\mathrm{d}\xi = [h_{\mathrm{T}} - \gamma_{\mathrm{T}}\xi]\,\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma_{\mathrm{T}}/\mu_{\mathrm{T}}}\mathrm{dW_T}, \tag{8.78}$$

$$\mathrm{d}\eta = [h_{\mathrm{P}} - \gamma_{\mathrm{P}}\eta]\,\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma_{\mathrm{P}}/\mu_{\mathrm{P}}}\mathrm{dW_P}, \tag{8.79}$$

where $\gamma_{\mathrm{T}},\ \gamma_{\mathrm{P}} \geq 0$ are two parameters governing the strength of the noise.

8.8.2 *Discretization of the NHL Piston*

Observations in Chap. 7 and our experience with the Nosé-Hoover-Langevin method lead us to propose splitting methods to separate the Ornstein-Uhlenbeck processes